

gradient descent and many others. It is similar to scikit-learn and NumPy in Python.

- **JavaCPP:** This is the bridge and interface tool for Java and C++. It provides the interface and APIs to access C++ code in Java without any intermediate or third-party application.
 - **DataVec:** This is the machine learning tool to extract, transform and load (ETL). The key function of DataVec is to transform the raw data to vector format with preprocessing to make it compatible with training in the ML implementations. It supports multiple formats including comma-separated values (CSVs), images, video, binary, text and many others.
 - **RL4J:** This refers to reinforcement learning for Java platforms with the implementation of Deep Q Learning, asynchronous actor-critic agents (A3C) and many other algorithms for reinforcement learning for JVM.
 - **Arbiter:** This is a tool to evaluate and test the algorithms of ML. It supports searching for hyper-parameters to extract the best fit configuration of neural networks.
- The following are the prerequisites to work with

DeepLearning4j:

- Java Development Kit (JDK)
- Eclipse IDE or IntelliJ IDEA
- Git
- Apache Maven

After installation of the required software tools, the examples and use cases from GitHub can be downloaded for

testing and further customisation of the code.

```
$ git clone https://github.com/deeplearning4j/dl4j-examples.git
$ cd dl4j-examples/
$ mvn clean install
```

From IntelliJ, the option of *Import Project* is selected, along with the directory 'dl4j-examples'.

Select 'Import project from external model' and Maven as shown in Figure 3.

Select the DeepLearning4j example for execution. In this example, *MLPClassifierLinear* is used and run. A series of scores will be displayed at the bottom.

Figure 6 depicts the graph to show the classified data using multi-layer perceptron (MLP).

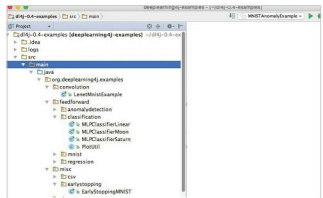


Figure 4: Select an example for execution

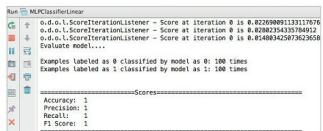


Figure 5: Output from executing the DeepLearning4j project

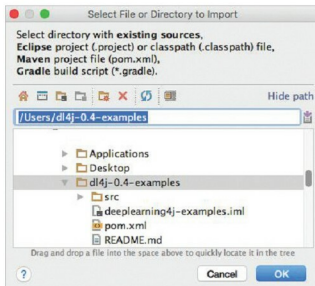


Figure 2: Import example project of DeepLearning4j

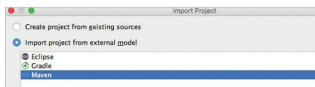


Figure 3: Import project and association with Maven

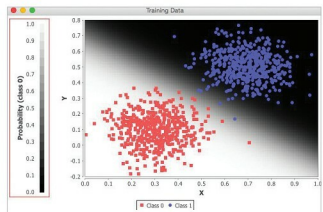


Figure 6: Graphical depiction of the classifier